

Data Mining

Chapter 3. Data Preprocessing

Esha Nawaz

Lecturer

The University of Faisalabad

- ❑ Data Preprocessing: An Overview
- ❑ Data Cleaning
- ❑ Data Integration
- ❑ Data Reduction and Transformation
- ❑ Dimensionality Reduction
- ❑ Summary

☐ Data cleaning

- ☐ Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies

☐ Data integration

- ☐ Integration of multiple databases, data cubes, or files

☐ Data reduction

- ☐ Dimensionality reduction
- ☐ Numerosity reduction
- ☐ Data compression

☐ Data transformation and data discretization

- ☐ Normalization
- ☐ Concept hierarchy generation

- ❑ Measures for data quality: A multidimensional view
 - ❑ Accuracy: correct or wrong, accurate or not
 - ❑ Completeness: not recorded, unavailable, ...
 - ❑ Consistency: some modified but some not, dangling, ...
 - ❑ Timeliness: timely update?
 - ❑ Believability: how trustable the data are correct?
 - ❑ Interpretability: how easily the data can be understood?

- ❑ Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, and transmission error
- ❑ Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - ❑ e.g., *Occupation* = “ ” (missing data)
- ❑ Noisy: containing noise, errors, or outliers
 - ❑ e.g., *Salary* = “-10” (an error)
- ❑ Inconsistent: containing discrepancies in codes or names, e.g.,
 - ❑ *Age* = “42”, *Birthday* = “03/07/2010”
 - ❑ Was rating “1, 2, 3”, now rating “A, B, C”
 - ❑ discrepancy between duplicate records
- ❑ Intentional (e.g., *disguised missing* data)
 - ❑ Jan. 1 as everyone’s birthday?

- ❑ Data is not always available
 - ❑ E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- ❑ Missing data may be due to
 - ❑ Equipment malfunction
 - ❑ Inconsistent with other recorded data and thus deleted
 - ❑ Data were not entered due to misunderstanding
 - ❑ Certain data may not be considered important at the time of entry
 - ❑ Did not register history or changes of the data
- ❑ Missing data may need to be inferred

- ❑ Ignore the tuple: usually done when class label is missing (when doing classification)
—not effective when the % of missing values per attribute varies considerably
- ❑ Fill in the missing value manually: tedious + infeasible?
- ❑ Fill in it automatically with
 - ❑ a global constant : e.g., “unknown”, a new class?!
 - ❑ the attribute mean
 - ❑ the attribute mean for all samples belonging to the same class: smarter
 - ❑ **the most probable value: inference-based such as Bayesian formula or decision tree**

- ❑ **Noise:** random error or variance in a measured variable
- ❑ **Incorrect attribute values** may be due to
 - ❑ Faulty data collection instruments
 - ❑ Data entry problems
 - ❑ Data transmission problems
 - ❑ Technology limitation
 - ❑ Inconsistency in naming convention
- ❑ **Other data problems**
 - ❑ Duplicate records
 - ❑ Incomplete data
 - ❑ Inconsistent data

❑ Binning

- ❑ First sort data and partition into (equal-frequency) bins
- ❑ Then one can **smooth by bin means, smooth by bin median, smooth by bin boundaries**, etc.

❑ Regression

- ❑ Smooth by fitting the data into regression functions

❑ Clustering

- ❑ Detect and remove outliers

❑ Semi-supervised: Combined computer and human inspection

- ❑ Detect suspicious values and check by human (e.g., deal with possible outliers)

❑ Data discrepancy detection

- ❑ Use metadata (e.g., domain, range, dependency, distribution)
- ❑ Check field overloading
- ❑ Check uniqueness rule, consecutive rule and null rule
- ❑ Use commercial tools
 - ❑ Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
 - ❑ Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

❑ Data migration and integration

- ❑ Data migration tools: allow transformations to be specified
- ❑ ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface
- ❑ Integration of the two processes
 - ❑ Iterative and interactive (e.g., Potter's Wheels)

- ❑ Data integration
 - ❑ Combining data from multiple sources into a coherent store
- ❑ Schema integration: e.g., $A.cust-id \equiv B.cust-\#$
 - ❑ Integrate metadata from different sources
- ❑ **Entity identification:**
 - ❑ Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- ❑ Detecting and resolving data value conflicts
 - ❑ For the same real world entity, attribute values from different sources are different
 - ❑ Possible reasons: different representations, different scales, e.g., metric vs. British units

- ❑ Redundant data occur often when integration of multiple databases
 - ❑ *Object identification*: The same attribute or object may have different names in different databases
 - ❑ *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- ❑ **Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis***
- ❑ Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Correlation Analysis (for Categorical Data)

□ χ^2 (chi-square) test:

$$\chi^2 = \sum_i^n \frac{\overset{\text{observed}}{\downarrow} (O_i - E_i)^2}{\underset{\text{expected}}{E_i}}$$

- Null hypothesis: The two distributions are independent
- The cells that contribute the most to the χ^2 value are those whose actual count is very different from the expected count
 - The larger the χ^2 value, the more likely the variables are related
- Note: Correlation does not imply causality
 - # of hospitals and # of car-theft in a city are correlated
 - Both are causally linked to the third variable: population

	Play chess	Not play chess	Sum (row)
Like science fiction	250 (90)	200 (360)	450
Not like science fiction	50 (210)	1000 (840)	1050
Sum(col.)	300	1200	1500

How to derive 90?
 $450/1500 * 300 = 90$

- χ^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

We can reject the null hypothesis of independence at a confidence level of 0.001

- It shows that like_science_fiction and play_chess are correlated in the group

Variance for Single Variable (Numerical Data)

- The variance of a random variable X provides a measure of how much the value of X deviates from the mean or expected value of X :

$$\sigma^2 = (1/n) \sum (x_i - \mu)^2$$

- where σ^2 is the variance of X , σ is called *standard deviation*
 μ is the mean, and μ = Mean of X
- That is, variance is the expected value of the square deviation from the mean
- Sample variance is the average squared deviation of the data value x_i from the sample mean $\hat{\mu}$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

Variance for Single Variable (Numerical Data)

Calculate the sample variance of the following data set:

3, 4, 6, 7, 7, 9, 13

$n = 7$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
3	-4	16
4	-3	9
6	-1	1
7	0	0
7	0	0
9	2	4
13	6	36
Total:		66

$$\bar{x} = \frac{3 + 4 + 6 + 7 + 7 + 9 + 13}{7}$$

$$\bar{x} = 7$$

$$66 \div 6 = 11$$

The variance = 11

- Covariance between two variables X_1 and X_2

$$\sigma_{12} = E[(X_1 - \mu_1)(X_2 - \mu_2)] = E[X_1 X_2] - \mu_1 \mu_2 = E[X_1 X_2] - E[X_1]E[X_2]$$

where $\mu_1 = E[X_1]$ is the respective mean or **expected value** of X_1 ; similarly for μ_2

- Sample covariance between X_1 and X_2 : $\hat{\sigma}_{12} = \frac{1}{n} \sum_{i=1}^n (x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)$

- Sample covariance is a generalization of the sample variance:

$$\hat{\sigma}_{11} = \frac{1}{n} \sum_{i=1}^n (x_{i1} - \hat{\mu}_1)(x_{i1} - \hat{\mu}_1) = \frac{1}{n} \sum_{i=1}^n (x_{i1} - \hat{\mu}_1)^2 = \hat{\sigma}_1^2$$

- **Positive covariance:** If $\sigma_{12} > 0$

- **Negative covariance:** If $\sigma_{12} < 0$

- **Independence:** If X_1 and X_2 are independent, $\sigma_{12} = 0$ but the reverse is not true

- Some pairs of random variables may have a covariance 0 but are not independent
- Only under some additional assumptions (e.g., the data follow multivariate normal distributions) does a covariance of 0 imply independence

Example: Calculation of Covariance

- Suppose two stocks X_1 and X_2 have the following values in one week:
 - (2, 5), (3, 8), (5, 10), (4, 11), (6, 14)
- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?

- Covariance formula

$$\sigma_{12} = E[(X_1 - \mu_1)(X_2 - \mu_2)] = E[X_1X_2] - \mu_1\mu_2 = E[X_1X_2] - E[X_1]E[X_2]$$

- Its computation can be simplified as: $\sigma_{12} = E[X_1X_2] - E[X_1]E[X_2]$

- $E(X_1) = (2 + 3 + 5 + 4 + 6) / 5 = 20/5 = 4$

- $E(X_2) = (5 + 8 + 10 + 11 + 14) / 5 = 48/5 = 9.6$

- $\sigma_{12} = (2 \times 5 + 3 \times 8 + 5 \times 10 + 4 \times 11 + 6 \times 14) / 5 - 4 \times 9.6 = 4$

- Thus, X_1 and X_2 rise together since $\sigma_{12} > 0$

Correlation between Two Numerical Variables

- ❑ **Correlation** between two variables X_1 and X_2 is the standard covariance, obtained by normalizing the covariance with the standard deviation of each variable

$$\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2} = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

- ❑ **Sample correlation** for two attributes X_1 and X_2 : $\hat{\rho}_{12} = \frac{\hat{\sigma}_{12}}{\hat{\sigma}_1 \hat{\sigma}_2} = \frac{\sum_{i=1}^n (x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)}{\sqrt{\sum_{i=1}^n (x_{i1} - \hat{\mu}_1)^2 \sum_{i=1}^n (x_{i2} - \hat{\mu}_2)^2}}$

where n is the number of tuples, μ_1 and μ_2 are the respective means of X_1 and X_2 , σ_1 and σ_2 are the respective standard deviation of X_1 and X_2

- ❑ If $\rho_{12} > 0$: A and B are positively correlated (X_1 's values increase as X_2 's)
 - ❑ The higher, the stronger correlation
- ❑ If $\rho_{12} = 0$: independent (under the same assumption as discussed in co-variance)
- ❑ If $\rho_{12} < 0$: negatively correlated

$$\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2} = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

→ Step 1: Find the Means (μ_1, μ_2)

- μ_1 (Mean of X_1): $(2 + 3 + 5 + 4 + 6)/5 = 20/5 = 4$
- μ_2 (Mean of X_2): $(5 + 8 + 10 + 11 + 14)/5 = 48/5 = 9.6$

→ Step 2: Calculate the Variances (σ_1^2, σ_2^2)

Variance is the average of the squared deviations from the mean ($E[(X - \mu)^2]$).

- σ_1^2 (Variance of X_1):

$$\frac{(2 - 4)^2 + (3 - 4)^2 + (5 - 4)^2 + (4 - 4)^2 + (6 - 4)^2}{5}$$

$$\frac{(-2)^2 + (-1)^2 + (1)^2 + (0)^2 + (2)^2}{5} = \frac{4 + 1 + 1 + 0 + 4}{5} = \frac{10}{5} = 2$$

- σ_2^2 (Variance of X_2):

$$\frac{(5 - 9.6)^2 + (8 - 9.6)^2 + (10 - 9.6)^2 + (11 - 9.6)^2 + (14 - 9.6)^2}{5}$$

$$\frac{(-4.6)^2 + (-1.6)^2 + (0.4)^2 + (1.4)^2 + (4.4)^2}{5} = \frac{21.16 + 2.56 + 0.16 + 1.96 + 19.36}{5} =$$

→ Step 3: Calculate the Covariance (σ_{12})

Covariance is the average of the product of deviations ($E[(X_1 - \mu_1)(X_2 - \mu_2)]$).

- σ_{12} :

$$\frac{(-2 \times -4.6) + (-1 \times -1.6) + (1 \times 0.4) + (0 \times 1.4) + (2 \times 4.4)}{5}$$

$$\frac{9.2 + 1.6 + 0.4 + 0 + 8.8}{5} = \frac{20}{5} = 4$$



✅ **Step 4: Final Correlation (ρ_{12})**

Now, we plug all three results into the formula from your image:

$$\rho_{12} = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

$$\rho_{12} = \frac{4}{\sqrt{2 \times 9.04}}$$

$$\rho_{12} = \frac{4}{\sqrt{18.08}}$$

$$\rho_{12} = \frac{4}{4.25205...}$$

$$\rho_{12} \approx 0.941$$



x_{i1}	x_{i2}	$(x_{i1} - \hat{\mu}_1)$	$(x_{i2} - \hat{\mu}_2)$	Product	$(x_{i1} - \hat{\mu}_1)^2$	$(x_{i2} - \hat{\mu}_2)^2$
2	5	-2	-4.6	9.2	4	21.16
3	8	-1	-1.6	1.6	1	2.56
5	10	1	0.4	0.4	1	0.16
4	11	0	1.4	0.0	0	1.96
6	14	2	4.4	8.8	4	19.36
Sum ($\sum$)				20.0	10	45.2



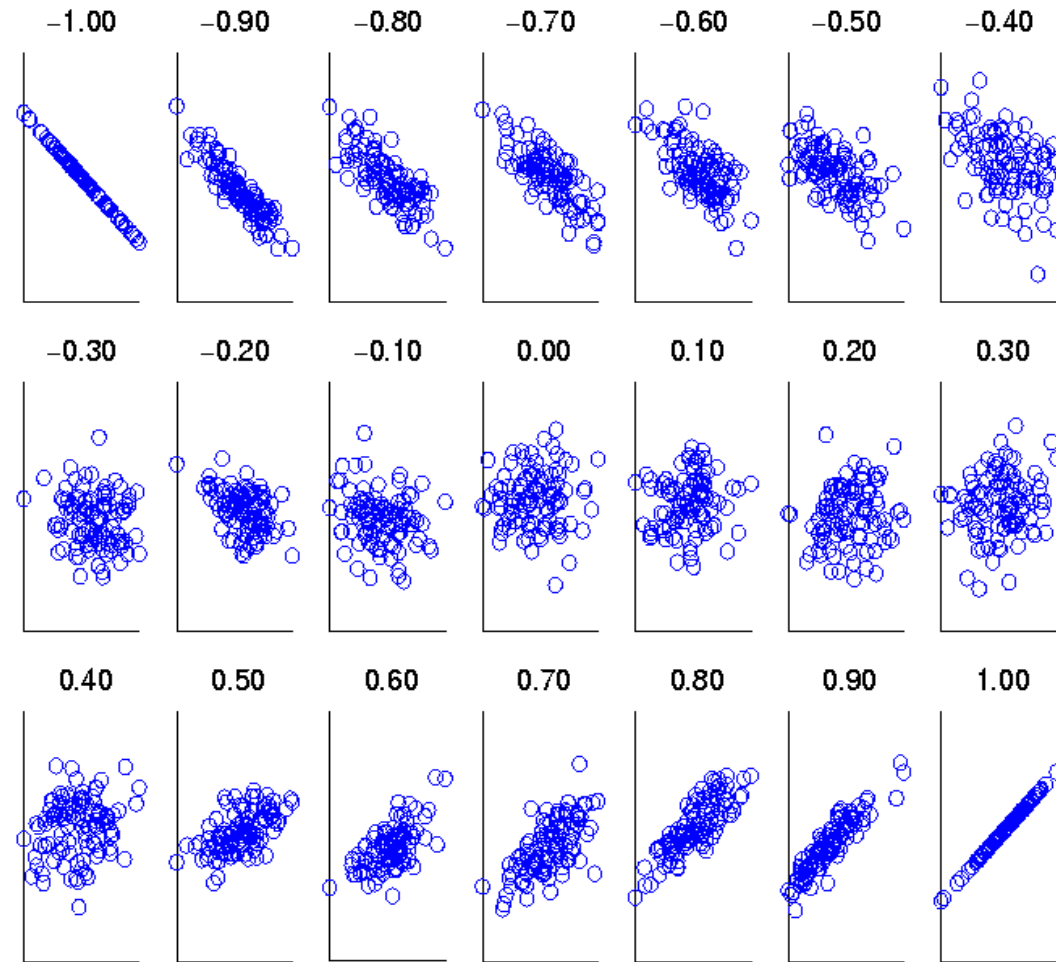
$$\hat{\rho}_{12} = \frac{\sum(x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)}{\sqrt{\sum(x_{i1} - \hat{\mu}_1)^2 \sum(x_{i2} - \hat{\mu}_2)^2}}$$

$$\hat{\rho}_{12} = \frac{20.0}{\sqrt{10 \times 45.2}}$$

$$\hat{\rho}_{12} = \frac{20.0}{\sqrt{452}}$$

$$\hat{\rho}_{12} = \frac{20.0}{21.260}$$

$$\hat{\rho}_{12} \approx 0.941$$



- Correlation coefficient value range: $[-1, 1]$
- A set of scatter plots shows sets of points and their correlation coefficients changing from -1 to 1

Figure 5.11. Scatter plots illustrating correlations from -1 to 1 .

- The variance and covariance information for the two variables X_1 and X_2 can be summarized as 2 X 2 covariance matrix as

$$\begin{aligned}\Sigma &= E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T] = E\left[\begin{pmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{pmatrix} \begin{pmatrix} X_1 - \mu_1 & X_2 - \mu_2 \end{pmatrix}\right] \\ &= \begin{pmatrix} E[(X_1 - \mu_1)(X_1 - \mu_1)] & E[(X_1 - \mu_1)(X_2 - \mu_2)] \\ E[(X_2 - \mu_2)(X_1 - \mu_1)] & E[(X_2 - \mu_2)(X_2 - \mu_2)] \end{pmatrix} \\ &= \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix}\end{aligned}$$

- Generalizing it to d dimensions, we have,

$$D = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{d1} & x_{d2} & \cdots & x_{dd} \end{pmatrix} \quad \Sigma = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T] = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_d^2 \end{pmatrix}$$

□ Data reduction:

- Obtain a reduced representation of the data set
 - much smaller in volume but yet produces *almost* the same analytical results

□ Why data reduction?—A database/data warehouse may store terabytes of data

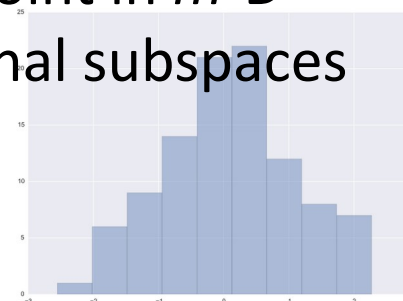
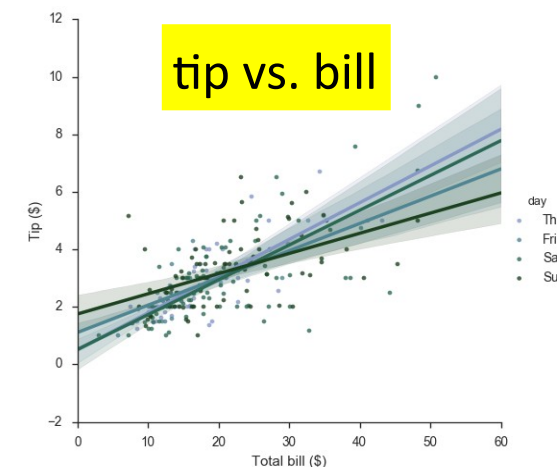
- Complex analysis may take a very long time to run on the complete data set

□ **Methods for data reduction** (also *data size reduction* or *numerosity reduction*)

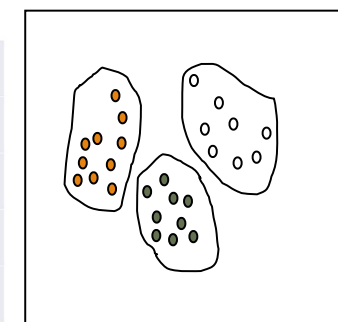
- Regression and Log-Linear Models
- Histograms, clustering, sampling
- Data cube aggregation
- Data compression

Data Reduction: Parametric vs. Non-Parametric Methods

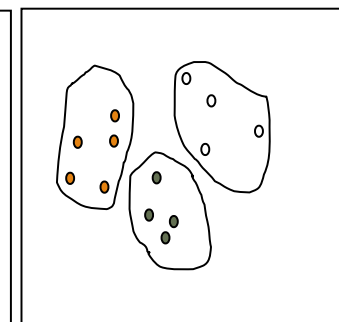
- ❑ Reduce data volume by choosing alternative, *smaller forms* of data representation
- ❑ **Parametric methods** (e.g., regression)
 - ❑ Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
 - ❑ Ex.: Log-linear models—obtain value at a point in m -D space as the product on appropriate marginal subspaces
- ❑ **Non-parametric methods**
 - ❑ Do not assume models
 - ❑ Major families: histograms, clustering, sampling.



Histogram

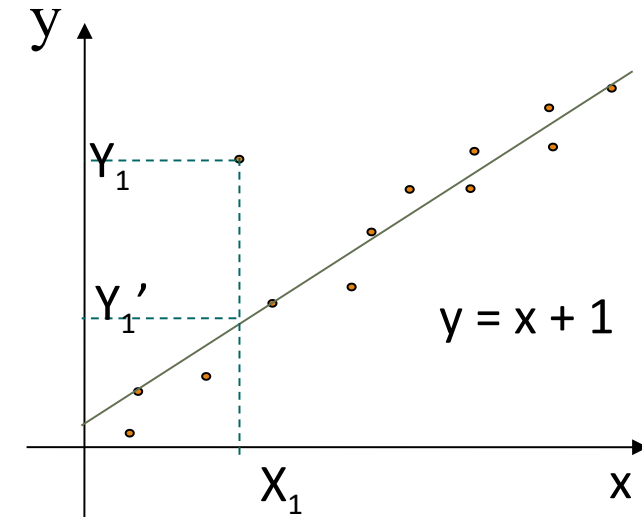


Clustering on the Raw Data



Stratified Sampling

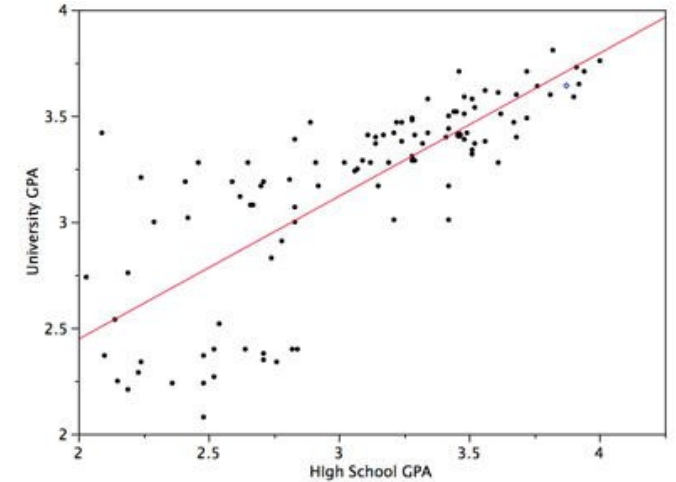
- ❑ Regression analysis: A collective name for techniques for the modeling and analysis of numerical data consisting of values of a **dependent variable** (also called **response variable** or *measurement*) and of one or more **independent variables** (also known as **explanatory variables** or **predictors**)
- ❑ The parameters are estimated so as to give a "**best fit**" of the data
- ❑ Most commonly the best fit is evaluated by using the **least squares method**, but other criteria have also been used



- ❑ Used for prediction (including forecasting of time-series data), inference, hypothesis testing, and modeling of causal relationships

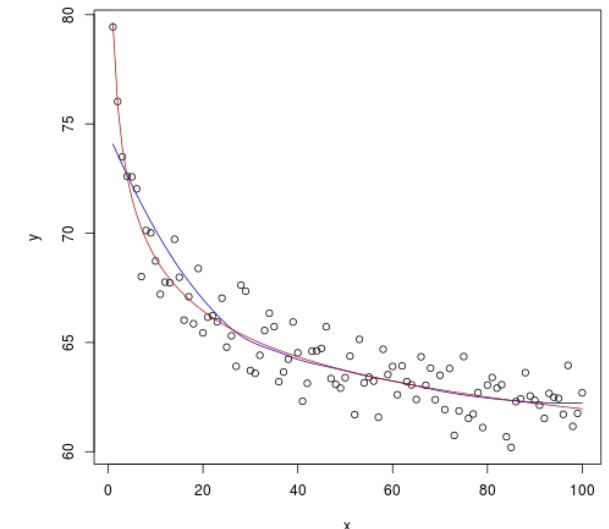
□ Linear regression: $Y = wX + b$

- Data modeled to fit a straight line
- Often uses the least-square method to fit the line
- Two regression coefficients, w and b , specify the line and are to be estimated by using the data at hand
- Using the least squares criterion to the known values of $Y_1, Y_2, \dots, X_1, X_2, \dots$



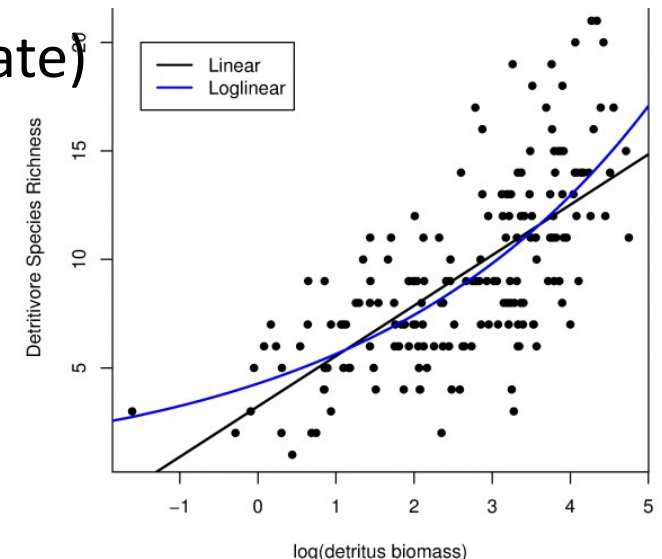
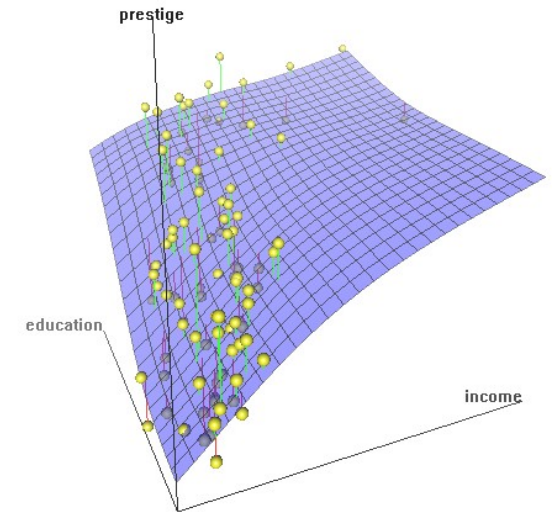
□ Nonlinear regression:

- Data are modeled by a function which is a nonlinear combination of the model parameters and depends on one or more independent variables
- The data are fitted by a method of successive approximations

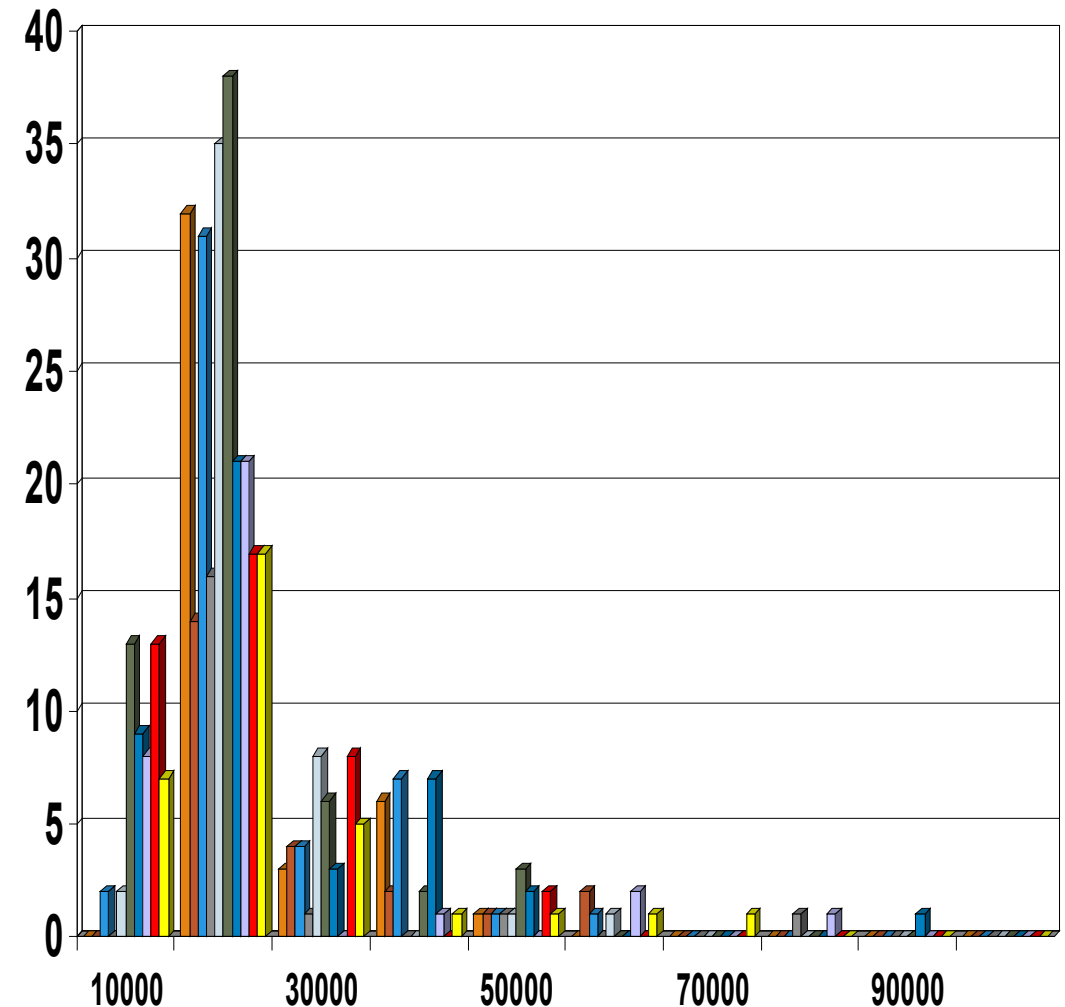


Multiple Regression and Log-Linear Models

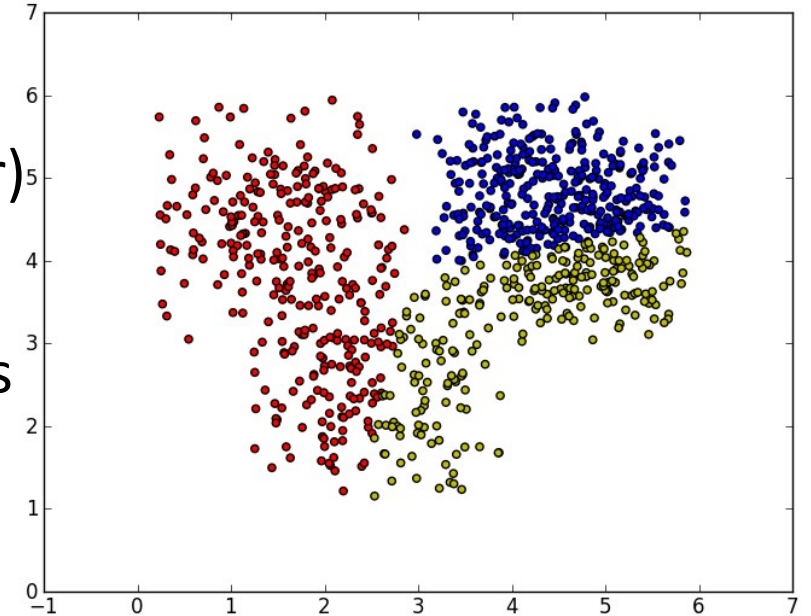
- ❑ Multiple regression: $Y = b_0 + b_1 X_1 + b_2 X_2$
 - ❑ Allows a response variable Y to be modeled as a linear function of multidimensional feature vector
 - ❑ Many nonlinear functions can be transformed into the above
- ❑ Log-linear model:
 - ❑ A math model that takes the form of a function whose logarithm is a linear combination of the parameters of the model, which makes it possible to apply (possibly multivariate) linear regression
 - ❑ Estimate the probability of each point (tuple) in a multi-dimen. space for a set of discretized attributes, based on a smaller subset of dimensional combinations
 - ❑ Useful for dimensionality reduction and data smoothing



- Divide data into buckets and store average (sum) for each bucket
- Partitioning rules:
 - Equal-width: equal bucket range
 - Equal-frequency (or equal-depth)



- Partition data set into clusters based on similarity, and store cluster representation (e.g., centroid and diameter) only
- Can be very effective if data is clustered but not if data is “smeared”
- Can have hierarchical clustering and be stored in multi-dimensional index tree structures
- There are many choices of clustering definitions and clustering algorithms
- Cluster analysis will be studied in depth in Chapter 10



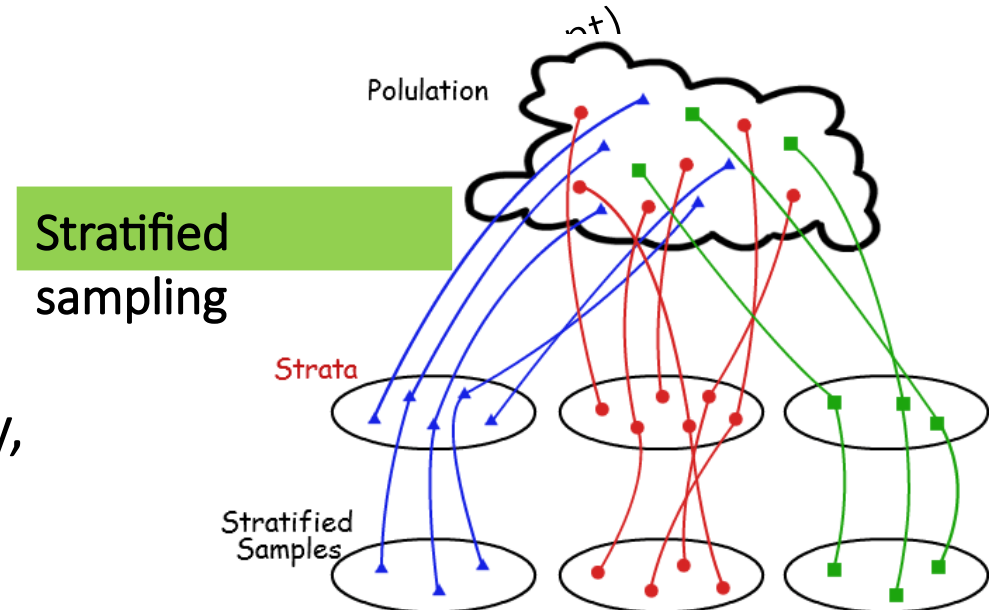
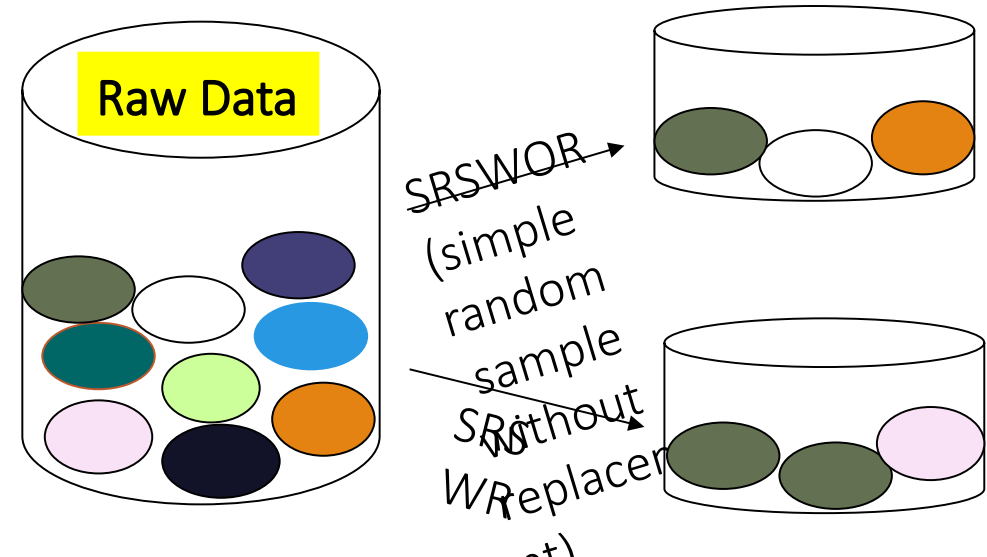
Announcement 1: Registration of the Overflow Session

- ❑ For those who would like to try to register our online session of the CS412 Fall classes
- ❑ Register before the end of Thursday (i.e., 11:59pm CT Thursday, Sept. 14)!
<http://go.cs.illinois.edu/CS412OverflowSection>
- ❑ Fill up all the needed information, waiting until the end of the week
- ❑ We will evaluate your application and decide if you can get into the overflown session

- ❑ Two choices, select one:
 - ❑ **Survey:** A focused, comprehensive survey on a focused topic of data mining
 - ❑ Ex. Clustering methods for heterogeneous information networks
 - ❑ **Research Project:** Working on a research problem related to data mining and write a paper/report, including a literature review, proposed method, datasets, and experiment results
- ❑ **Grouping:** Single person or a group of maximum 3 students
- ❑ **Submission on Compass2g**
 - ❑ One page proposal @the end of 4th week: Title, abstract, datasets, references
 - ❑ Final report: similar to a conference paper, usually 10-12 pages in double columns
- ❑ **Piazza:** Containing more detailed description, please raise questions on Piazza

- ❑ Sampling: obtaining a small sample s to represent the whole data set N
- ❑ Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- ❑ Key principle: Choose a **representative** subset of the data
 - ❑ Simple random sampling may have very poor performance in the presence of skew
 - ❑ Develop adaptive sampling methods, e.g., stratified sampling:
- ❑ Note: Sampling may not reduce database I/Os (page at a time)

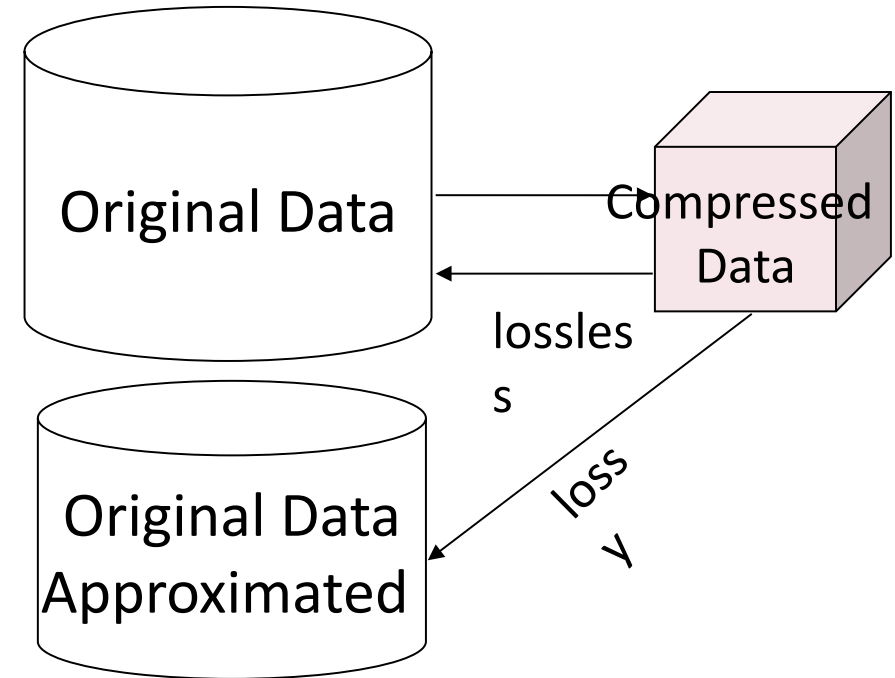
- ❑ **Simple random sampling:** equal probability of selecting any particular item
- ❑ **Sampling without replacement**
 - ❑ Once an object is selected, it is removed from the population
- ❑ **Sampling with replacement**
 - ❑ A selected object is not removed from the population
- ❑ **Stratified sampling**
 - ❑ Partition (or cluster) the data set, and draw samples from each partition (proportionally, i.e., approximately the same percentage of the data)



- The lowest level of a data cube (base cuboid)
 - The aggregated data for an **individual entity of interest**
 - E.g., a customer in a phone calling data warehouse
- Multiple levels of aggregation in data cubes
 - Further reduce the size of data to deal with
- Reference appropriate levels
 - Use the smallest representation which is enough to solve the task
- Queries regarding aggregated information should be answered using data cube, when possible



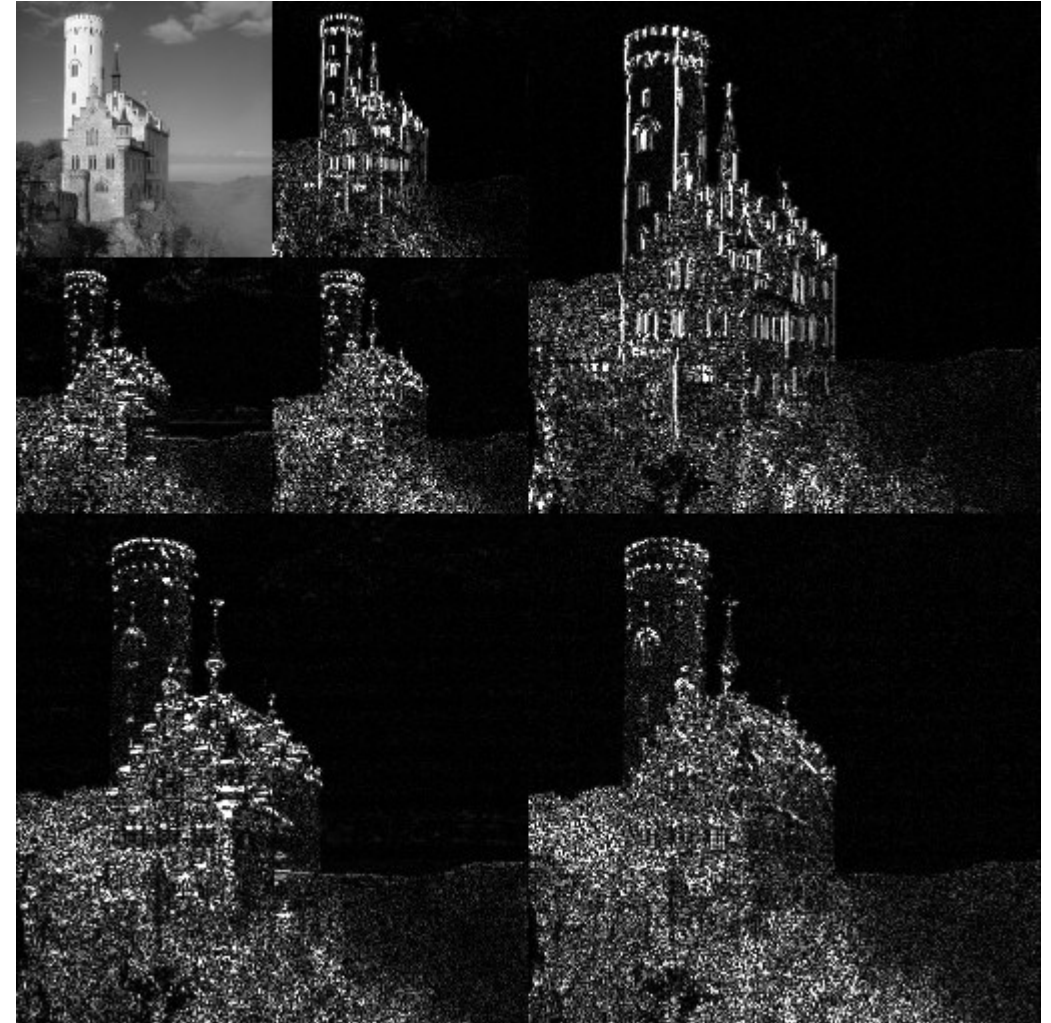
- ❑ String compression
 - ❑ There are extensive theories and well-tuned algorithms
 - ❑ Typically lossless, but only limited manipulation is possible without expansion
- ❑ Audio/video compression
 - ❑ Typically lossy compression, with progressive refinement
 - ❑ Sometimes small fragments of signal can be reconstructed without reconstructing the whole
- ❑ Time sequence is not audio
 - ❑ Typically short and vary slowly with time
- ❑ Data reduction and dimensionality reduction may also be considered as forms of data compression



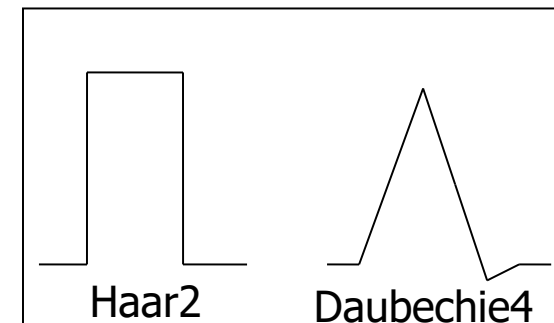
Lossy vs. lossless compression

Wavelet Transform: A Data Compression Technique

- ❑ Wavelet Transform
 - ❑ Decomposes a signal into different frequency subbands
 - ❑ Applicable to n-dimensional signals
- ❑ Data are transformed to preserve relative distance between objects at different levels of resolution
- ❑ Allow natural clusters to become more distinguishable
- ❑ Used for image compression



- ❑ Discrete wavelet transform (DWT) for linear signal processing, multi-resolution analysis
- ❑ Compressed approximation: Store only a small fraction of the strongest of the wavelet coefficients
- ❑ Similar to discrete Fourier transform (DFT), but better lossy compression, localized in space
- ❑ Method:
 - ❑ Length, L , must be an integer power of 2 (padding with 0's, when necessary)
 - ❑ Each transform has 2 functions: smoothing, difference
 - ❑ Applies to pairs of data, resulting in two set of data of length $L/2$
 - ❑ Applies two functions recursively, until reaches the desired length



- ❑ Wavelets: A math tool for space-efficient hierarchical decomposition of functions
- ❑ $S = [2, 2, 0, 2, 3, 5, 4, 4]$ can be transformed to $S_{\wedge} = [2^{3/4}, -1^{1/4}, 1/2, 0, 0, -1, -1, 0]$
- ❑ Compression: many small detail coefficients can be replaced by 0's, and only the significant coefficients are retained

Resolution	Averages	Detail Coefficients
8	$[2, 2, 0, 2, 3, 5, 4, 4]$	
4	$[2, 1, 4, 4]$	$[0, -1, -1, 0]$
2	$[1\frac{1}{2}, 4]$	$[\frac{1}{2}, 0]$
1	$[2\frac{3}{4}]$	$[-1\frac{1}{4}]$

- ❑ Use hat-shape filters
 - ❑ Emphasize region where points cluster
 - ❑ Suppress weaker information in their boundaries
- ❑ Effective removal of outliers
 - ❑ Insensitive to noise, insensitive to input order
- ❑ Multi-resolution
 - ❑ Detect arbitrary shaped clusters at different scales
- ❑ Efficient
 - ❑ Complexity $O(N)$
- ❑ Only applicable to low dimensional data

- ❑ A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values
- ❑ Methods
 - ❑ Smoothing: Remove noise from data
 - ❑ Attribute/feature construction
 - ❑ New attributes constructed from the given ones
 - ❑ Aggregation: Summarization, data cube construction
 - ❑ Normalization: Scaled to fall within a smaller, specified range
 - ❑ min-max normalization
 - ❑ z-score normalization
 - ❑ normalization by decimal scaling
 - ❑ Discretization: Concept hierarchy climbing

□ Min-max normalization: to $[new_min_A, new_max_A]$

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

□ Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]

□ Then \$73,000 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

□ Z-score normalization (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

Z-score: The distance between the raw score and the population mean in the unit of the standard deviation

□ Ex. Let $\mu = 54,000$, $\sigma = 16,000$. Then $\frac{73,600 - 54,000}{16,000} = 1.225$

□ Normalization by decimal scaling

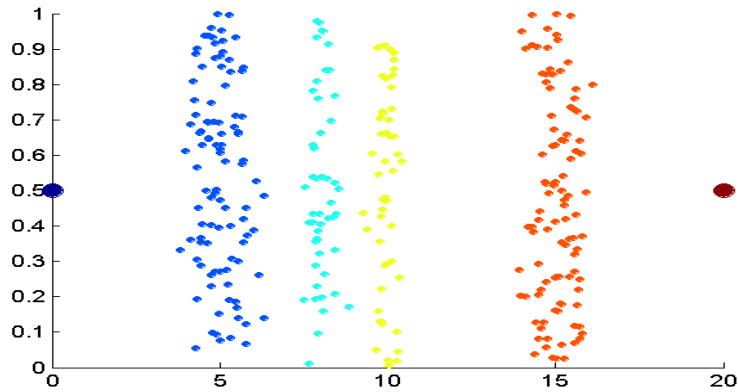
$$v' = \frac{v}{10^j} \quad \text{Where } j \text{ is the smallest integer such that } \text{Max}(|v'|) < 1$$

- ❑ Three types of attributes
 - ❑ Nominal—values from an unordered set, e.g., color, profession
 - ❑ Ordinal—values from an ordered set, e.g., military or academic rank
 - ❑ Numeric—real numbers, e.g., integer or real numbers
- ❑ Discretization: Divide the range of a continuous attribute into intervals
 - ❑ Interval labels can then be used to replace actual data values
 - ❑ Reduce data size by discretization
 - ❑ Supervised vs. unsupervised
 - ❑ Split (top-down) vs. merge (bottom-up)
 - ❑ Discretization can be performed recursively on an attribute
 - ❑ Prepare for further analysis, e.g., classification

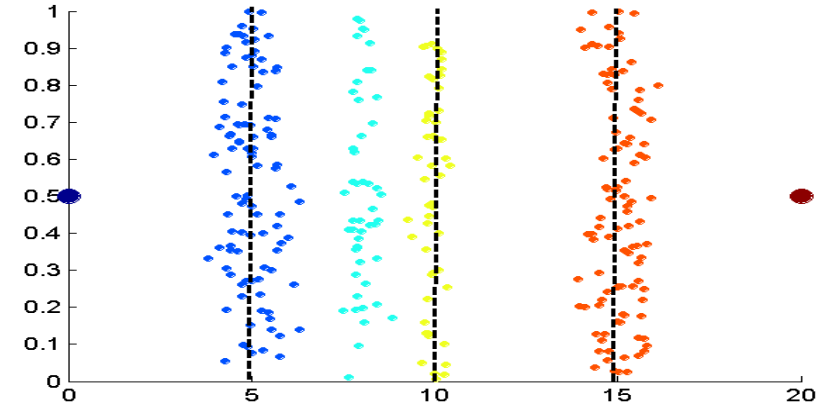
- ❑ Binning
 - ❑ Top-down split, unsupervised
- ❑ Histogram analysis
 - ❑ Top-down split, unsupervised
- ❑ Clustering analysis
 - ❑ Unsupervised, top-down split or bottom-up merge
- ❑ Decision-tree analysis
 - ❑ Supervised, top-down split
- ❑ Correlation (e.g., χ^2) analysis
 - ❑ Unsupervised, bottom-up merge
- ❑ Note: All the methods can be applied recursively

- ❑ **Equal-width** (distance) partitioning
 - ❑ Divides the range into N intervals of equal size: uniform grid
 - ❑ if A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A)/N$.
 - ❑ The most straightforward, but outliers may dominate presentation
 - ❑ Skewed data is not handled well
- ❑ **Equal-depth** (frequency) partitioning
 - ❑ Divides the range into N intervals, each containing approximately same number of samples
 - ❑ Good data scaling
 - ❑ Managing categorical attributes can be tricky

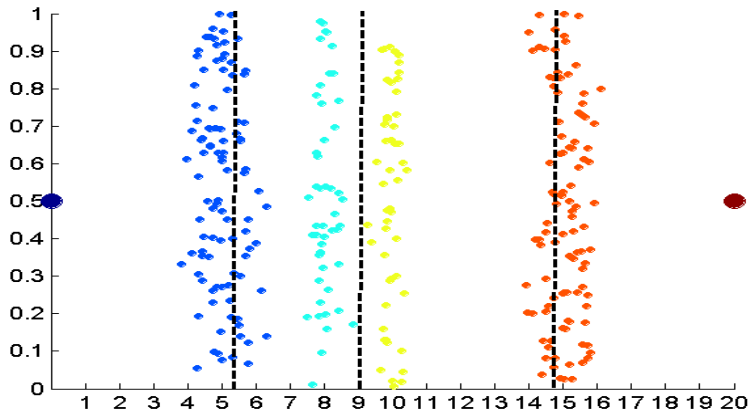
- ❑ Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
- * Partition into equal-frequency (**equi-depth**) bins:
 - Bin 1: 4, 8, 9, 15
 - Bin 2: 21, 21, 24, 25
 - Bin 3: 26, 28, 29, 34
- * Smoothing by **bin means**:
 - Bin 1: 9, 9, 9, 9
 - Bin 2: 23, 23, 23, 23
 - Bin 3: 29, 29, 29, 29
- * Smoothing by **bin boundaries**:
 - Bin 1: 4, 4, 4, 15
 - Bin 2: 21, 21, 25, 25
 - Bin 3: 26, 26, 26, 34



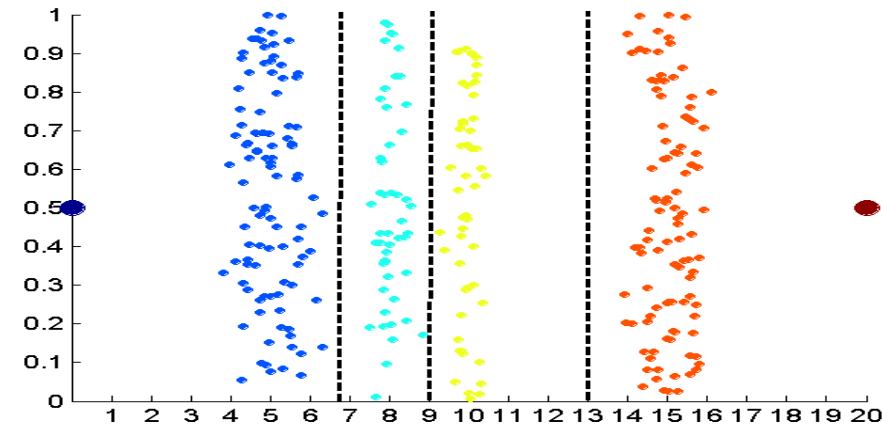
Data



Equal width (distance) binning



Equal depth (frequency) (binning)



K-means clustering leads to better results

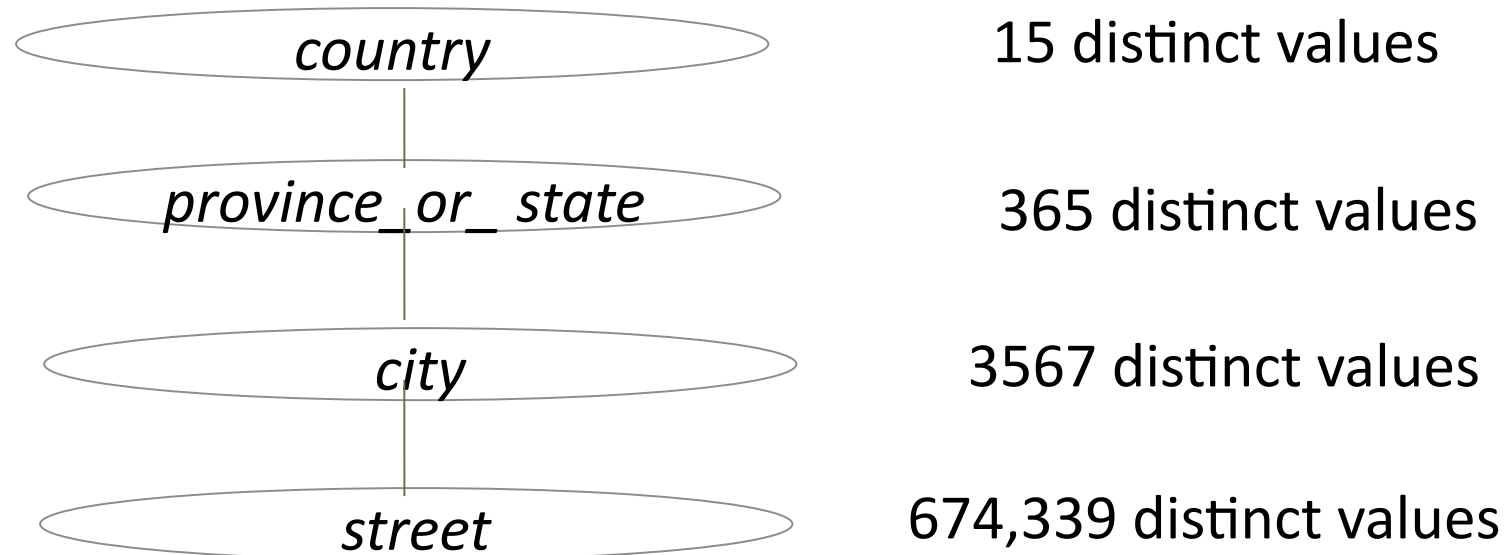
- ❑ Classification (e.g., decision tree analysis)
 - ❑ Supervised: Given class labels, e.g., cancerous vs. benign
 - ❑ Using *entropy* to determine split point (discretization point)
 - ❑ Top-down, recursive split
 - ❑ Details to be covered in Chapter “Classification”
- ❑ Correlation analysis (e.g., Chi-merge: χ^2 -based discretization)
 - ❑ Supervised: use class information
 - ❑ Bottom-up merge: Find the best neighboring intervals (those having similar distributions of classes, i.e., low χ^2 values) to merge
 - ❑ Merge performed recursively, until a predefined stopping condition

- ❑ **Concept hierarchy** organizes concepts (i.e., attribute values) hierarchically and is usually associated with each dimension in a data warehouse
- ❑ Concept hierarchies facilitate drilling and rolling in data warehouses to view data in multiple granularity
- ❑ Concept hierarchy formation: Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for *age*) by higher level concepts (such as *youth, adult, or senior*)
- ❑ Concept hierarchies can be explicitly specified by domain experts and/or data warehouse designers
- ❑ Concept hierarchy can be automatically formed for both numeric and nominal data—For numeric data, use discretization methods shown

Concept Hierarchy Generation for Nominal Data

- ❑ Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts
 - ❑ *street < city < state < country*
- ❑ Specification of a hierarchy for a set of values by explicit data grouping
 - ❑ {Urbana, Champaign, Chicago} < Illinois
- ❑ Specification of only a partial set of attributes
 - ❑ E.g., only *street < city*, not others
- ❑ Automatic generation of hierarchies (or attribute levels) by the analysis of the number of distinct values
 - ❑ E.g., for a set of attributes: {*street, city, state, country*}

- ❑ Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
 - ❑ The attribute with the most distinct values is placed at the lowest level of the hierarchy
 - ❑ Exceptions, e.g., weekday, month, quarter, year



❑ **Curse of dimensionality**

- ❑ When dimensionality increases, data becomes increasingly sparse
- ❑ Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
- ❑ The possible combinations of subspaces will grow exponentially

❑ **Dimensionality reduction**

- ❑ Reducing the number of random variables under consideration, via obtaining a set of principal variables

❑ **Advantages of dimensionality reduction**

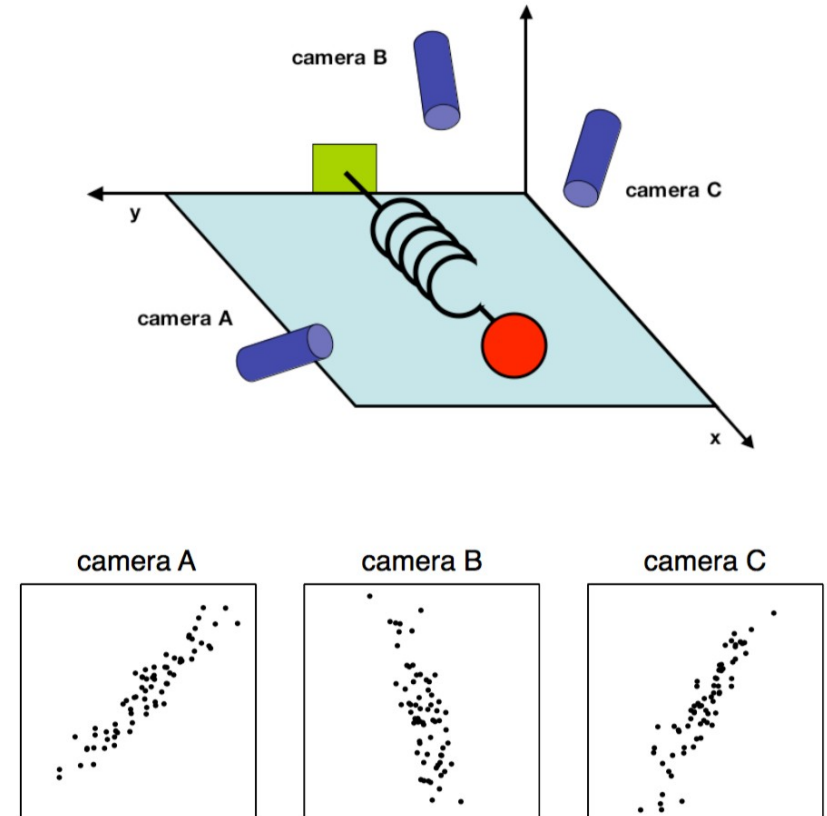
- ❑ Avoid the curse of dimensionality
- ❑ Help eliminate irrelevant features and reduce noise
- ❑ Reduce time and space required in data mining
- ❑ Allow easier visualization

- ❑ Dimensionality reduction methodologies
 - ❑ **Feature selection:** Find a subset of the original variables (or features, attributes)
 - ❑ **Feature extraction:** Transform the data in the high-dimensional space to a space of fewer dimensions
- ❑ Some typical dimensionality methods
 - ❑ Principal Component Analysis
 - ❑ Supervised and nonlinear techniques
 - ❑ Feature subset selection
 - ❑ Feature creation



Principal Component Analysis (PCA)

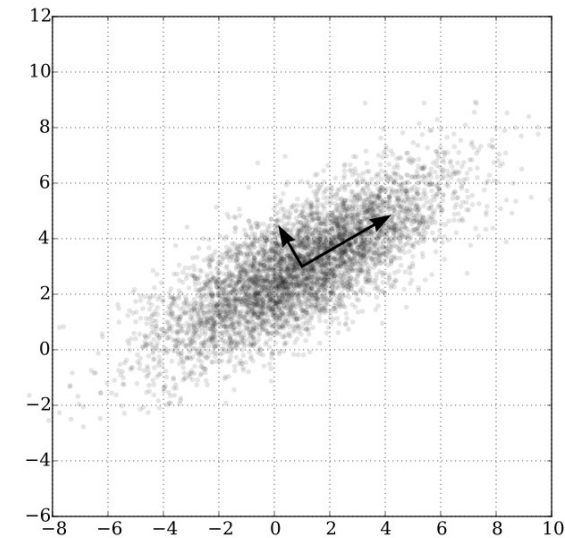
- ❑ PCA: A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called ***principal components***
- ❑ The original data are projected onto a much smaller space, resulting in dimensionality reduction
- ❑ Method: Find the eigenvectors of the covariance matrix, and these eigenvectors define the new space



Ball travels in a straight line. Data from three cameras contain much redundancy

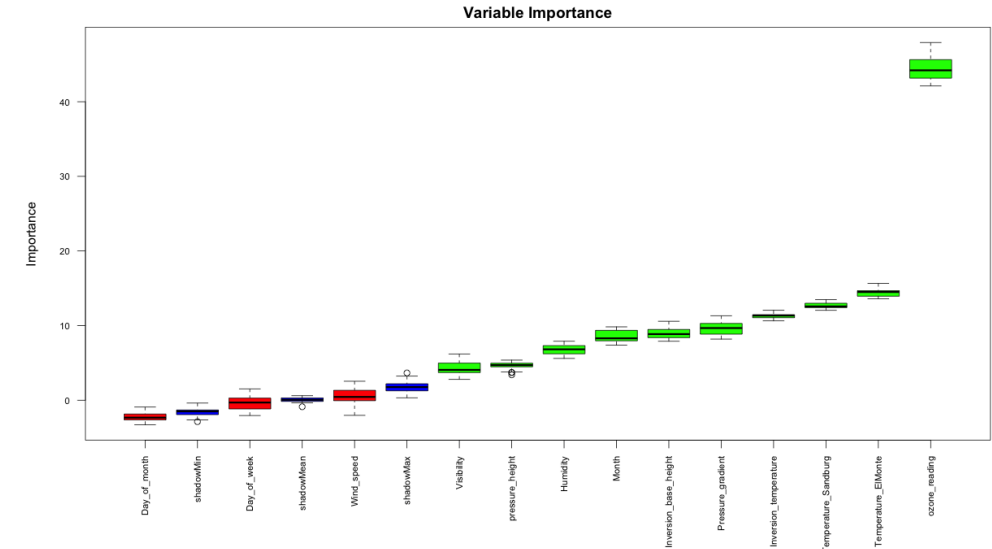
Principal Component Analysis (Method)

- Given N data vectors from n -dimensions, find $k \leq n$ orthogonal vectors (*principal components*) best used to represent data
 - Normalize input data: Each attribute falls within the same range
 - Compute k orthonormal (unit) vectors, i.e., *principal components*
 - Each input data (vector) is a linear combination of the k principal component vectors
 - The principal components are sorted in order of decreasing “significance” or strength
 - Since the components are sorted, the size of the data can be reduced by eliminating the *weak components*, i.e., those with low variance (i.e., using the strongest principal components, to reconstruct a good approximation of the original data)
- Works for numeric data only



Ack. Wikipedia: Principal Component Analysis

- ❑ Another way to reduce dimensionality of data
- ❑ Redundant attributes
 - ❑ Duplicate much or all of the information contained in one or more other attributes
 - ❑ E.g., purchase price of a product and the amount of sales tax paid
- ❑ Irrelevant attributes
 - ❑ Contain no information that is useful for the data mining task at hand
 - ❑ Ex. A student's ID is often irrelevant to the task of predicting his/her GPA



- ❑ There are 2^d possible attribute combinations of d attributes
- ❑ Typical heuristic attribute selection methods:
 - ❑ Best single attribute under the attribute independence assumption: choose by significance tests
 - ❑ Best step-wise feature selection:
 - ❑ The best single-attribute is picked first
 - ❑ Then next best attribute condition to the first, ...
 - ❑ Step-wise attribute elimination:
 - ❑ Repeatedly eliminate the worst attribute
 - ❑ Best combined attribute selection and elimination
 - ❑ Optimal branch and bound:
 - ❑ Use attribute elimination and backtracking

- ❑ Create new attributes (features) that can capture the important information in a data set more effectively than the original ones
- ❑ Three general methodologies
 - ❑ Attribute extraction
 - ❑ Domain-specific
 - ❑ Mapping data to new space (see: data reduction)
 - ❑ E.g., Fourier transformation, wavelet transformation, manifold approaches (not covered)
 - ❑ Attribute construction
 - ❑ Combining features (see: discriminative frequent patterns in Chapter on “Advanced Classification”)
 - ❑ Data discretization

- ❑ **Data quality:** accuracy, completeness, consistency, timeliness, believability, interpretability
- ❑ **Data cleaning:** e.g. missing/noisy values, outliers
- ❑ **Data integration** from multiple sources:
 - ❑ Entity identification problem; Remove redundancies; Detect inconsistencies
- ❑ **Data reduction, data transformation and data discretization**
 - ❑ Numerosity reduction; Data compression
 - ❑ Normalization; Concept hierarchy generation
- ❑ **Dimensionality reduction**

- ❑ D. P. Ballou and G. K. Tayi. Enhancing data quality in data warehouse environments. Comm. of ACM, 42:73-78, 1999
- ❑ T. Dasu and T. Johnson. Exploratory Data Mining and Data Cleaning. John Wiley, 2003
- ❑ T. Dasu, T. Johnson, S. Muthukrishnan, V. Shkapenyuk. [Mining Database Structure; Or, How to Build a Data Quality Browser](#). SIGMOD'02
- ❑ H. V. Jagadish et al., Special Issue on Data Reduction Techniques. Bulletin of the Technical Committee on Data Engineering, 20(4), Dec. 1997
- ❑ D. Pyle. Data Preparation for Data Mining. Morgan Kaufmann, 1999
- ❑ E. Rahm and H. H. Do. Data Cleaning: Problems and Current Approaches. *IEEE Bulletin of the Technical Committee on Data Engineering*. Vol.23, No.4
- ❑ V. Raman and J. Hellerstein. Potters Wheel: An Interactive Framework for Data Cleaning and Transformation, VLDB'2001
- ❑ T. Redman. Data Quality: Management and Technology. Bantam Books, 1992
- ❑ R. Wang, V. Storey, and C. Firth. A framework for analysis of data quality research. IEEE Trans. Knowledge and Data Engineering, 7:623-640, 1995